International Trade[1]

Lecture Note: Review of the CES Demand

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[1] The notes are still preliminary and in beta. If you find any typo or mistake, please send it to malfaro@ualberta.ca.

Contents

1 Introduction **1**

2 Setup **1**

2.1 Optimal Choices 2

3 CES as Representation of Differentiated Goods **3**

3.1 Horizontal Differentiation 3

3.2 Vertical Differentiation 6

4 Price Elasticity of Demand **8**

5 Welfare **9**

5.1 Welfare Determinants 10

6 A Continuum of Goods **12**

7 CES from a Random Utility Model (OPTIONAL) **13**

Notation

This is a derivation

This is some comment

This is a comment on advanced topics which are not part of the course (you can ignore it without loss of continuity regarding the text)

- The symbol ":=" means "by definition".
- I denote vectors by bold lowercase letters (for instance, $\mathbf{x}$) and matrices by bold capital letters (for instance, $\mathbf{X}$).
- To differentiate between the *verb* "maximize" and the *operator* "maximum", I denote the former with "max" and the latter with "sup" (i.e., supremum). The same caveat applies to "minimize" and "minimum", where I use "min" and "inf" respectively, with the latter indicating infimum.
- "iff" means "if and only if"
- $\exp(x)$ is equivalent to e^x.
- Random variables are denoted with a bar below. For instance, $\underline{x}$.

The notes contain hyperlinks in both blue and red text, allowing you to quickly navigate to specific sections of the document. Most PDF readers offer a keyboard shortcut that takes you back to the original text location (e.g., Alt+Left Arrow on Acrobat Reader).

1 Introduction

This note introduces the **constant elasticity of substitution** (**CES**) demand. This demand is widely used across several fields in Economics, serving as the standard demand in International Trade and Macroeconomics. Its popularity stems from its high tractability for theoretical and empirical analysis. Additionally, the CES system embodies homothetic preferences, a property that simplifies analysis.[1]

2 Setup

We consider an industry consisting of a differentiated good. This good comprises a discrete set of varieties $\Omega := \{1, 2, ..., M\}$, where each variety $\omega \in \Omega$ is characterized by its price p_ω and a parameter $z_\omega > 0$ referred to as variety ω's appeal. While some authors directly refer to z_ω as quality, appeal is a broader concept: it captures any non-price attribute of ω (objective or subjective) that increases the attractiveness of a variety.

Demand in the industry derives from the decisions of a representative consumer, who allocates an exogenous expenditure y to the differentiated good. The consumer's preferences are represented by a CES utility function, formally given by

$$U\left[(q_\omega)_{\omega\in\Omega}\right] := \left[\sum_{\omega\in\Omega} (z_\omega)^{\frac{\delta}{\sigma}} (q_\omega)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}, \tag{1}$$

where $\delta > 0$, $\sigma > 1$, and q_ω is the quantity demanded of ω. The parameter σ is known as the **elasticity of substitution**. The condition $\sigma > 1$ ensures that varieties are imperfect substitutes, meaning that consumers are willing to substitute across varieties, but not in a one-to-one manner.

The CES encompasses several special cases, depending on the value of σ:

- If $\sigma \to \infty$, then U converges to the linear utility function.
- If $\sigma \to 0$, then U converges to the Leontief utility function (perfect complements).
- If $\sigma \to 1$, then U converges to the Cobb Douglas utility function.

[1] In empirical research, two demand systems are especially common: the CES and the Logit. The Logit model, primarily employed in Industrial Organization, shares several structural similarities with the CES. For this reason, a solid understanding of the CES system provides a strong foundation for working with the Logit model as well.

The standard representation of CES assumes that all varieties share the same appeal parameter. In particular, $z_\omega = 1$ for all $\omega \in \Omega$. Moreover, since any monotone transformation of a utility function preserves the underlying preferences, it is common to find alternative but equivalent representations for the CES utility function. For instance, under the assumption that $z_\omega = 1$ for all $\omega \in \Omega$, the CES is usually presented as:

$$U\left[(q_\omega)_{\omega\in\Omega}\right] := \sum_{\omega\in\Omega} q_j^\rho,$$

$$U\left[(q_\omega)_{\omega\in\Omega}\right] := \left(\sum_{\omega\in\Omega} q_j^\rho\right)^{\frac{1}{\rho}}.$$

Note that the latter functional form is equivalent to (1) by setting $\rho := \frac{\sigma-1}{\sigma}$.
In this note, we express the CES demand in terms of the parameter σ, as it provides a more direct interpretation of the results in terms of the elasticity of substitution.

2.1 Optimal Choices

The consumer's optimization problem is

$$\max_{(q_\omega)_{\omega\in\Omega}} U\left[(q_\omega)_{\omega\in\Omega}\right] = \left[\sum_{\omega\in\Omega} (z_\omega)^{\frac{\delta}{\sigma}} (q_\omega)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \text{ subject to } y = \sum_{\omega\in\Omega} p_\omega q_\omega. \tag{2}$$

Solving this problem, the optimal quantity demanded of variety ω is

$$q_\omega = y\frac{(z_\omega)^\delta (p_\omega)^{-\sigma}}{\mathbb{P}^{1-\sigma}}, \tag{3}$$

where $\mathbb{P}$ is a price index defined by

$$\mathbb{P} := \left[\sum_{\omega\in\Omega} (z_\omega)^\delta (p_\omega)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}. \tag{4}$$

The CES utility function has no corner solutions and has a unique interior solution. To characterize the solution through the first-order conditions, note that maximizing $\left[\sum_{\omega\in\Omega} (z_\omega)^{\frac{\delta}{\sigma}} (q_\omega)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}$ is equivalent to maximizing $\sum_{\omega\in\Omega} (z_\omega)^{\frac{\delta}{\sigma}} (q_\omega)^{\frac{\sigma-1}{\sigma}}$. Hence, the Lagrangian is
$\mathscr{L} = \sum_{\omega\in\Omega} (z_\omega)^{\frac{\delta}{\sigma}} (q_\omega)^{\frac{\sigma-1}{\sigma}} + \lambda\left(Y - \sum_{\omega\in\Omega} p_\omega q_\omega\right)$,
where λ is the Lagrange multiplier.
The first-order condition for variety ω is
$(z_\omega)^{\frac{\delta}{\sigma}} \frac{\sigma-1}{\sigma} (q_\omega)^{\frac{-1}{\sigma}} - \lambda p_\omega = 0$
$\Rightarrow \frac{\sigma-1}{\sigma} (q_\omega)^{\frac{-1}{\sigma}} = -\lambda p_\omega (z_\omega)^{-\frac{\delta}{\sigma}}$
$\Rightarrow \left(\frac{\sigma-1}{\sigma}\right)^{-\sigma} q_\omega = -(\lambda)^{-\sigma} (p_\omega)^{-\sigma} (z_\omega)^\delta$
where we have simplified the notation by directly referring to the optimal solution as q_ω.
There are M first-order conditions, one for each $\omega \in \Omega$. Take $\omega', \omega'' \in \Omega$, and divide their corresponding first-order conditions:

$$\frac{q_{\omega'}}{q_{\omega''}} = \left(\frac{z_{\omega'}}{z_{\omega''}}\right)^{\delta} \left(\frac{p_{\omega'}}{p_{\omega''}}\right)^{-\sigma} \Rightarrow \frac{p_{\omega'} q_{\omega'}}{p_{\omega''} q_{\omega''}} = \left(\frac{z_{\omega'}}{z_{\omega''}}\right)^{\delta} \left(\frac{p_{\omega'}}{p_{\omega''}}\right)^{1-\sigma}$$

Now, fix ω'' and sum over ω',

$\sum_{\omega' \in \Omega} \frac{p_{\omega'} q_{\omega'}}{p_{\omega''} q_{\omega''}} = \sum_{\omega' \in \Omega} \left(\frac{p_{\omega'}}{p_{\omega''}}\right)^{1-\sigma}$

$\Rightarrow \frac{1}{p_{\omega''} q_{\omega''}} \sum_{\omega' \in \Omega} p_{\omega'} q_{\omega'} = \left(\frac{1}{z_{\omega''}}\right)^{\delta} \left(\frac{1}{p_{\omega''}}\right)^{1-\sigma} \sum_{\omega' \in \Omega} (z_{\omega'})^{\delta} (p_{\omega'})^{1-\sigma}$.

Using that $y = \sum_{\omega' \in \Omega} p_{\omega'} q_{\omega'}$ and defining $\mathbb{P} := \left[\sum_{\omega' \in \Omega} (z_{\omega'})^{\delta} (p_{\omega'})^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$, we get $q_{\omega} = y (z_{\omega})^{\delta} (p_{\omega})^{-\sigma} \mathbb{P}^{\sigma-1}$.

Likewise, the expenditure on variety ω is defined by $r_{\omega} := p_{\omega} q_{\omega}$, where r stands for revenue. It is given by

$$r_{\omega} = y \frac{(z_{\omega})^{\delta} (p_{\omega})^{1-\sigma}}{\mathbb{P}^{1-\sigma}}.$$

This expression allows us to identify ω's market share, which is formally defined by $s_{\omega} := \frac{y_{\omega}}{y}$ and given by

$$\begin{aligned} s_{\omega} &= \frac{(z_{\omega})^{\delta} (p_{\omega})^{1-\sigma}}{\mathbb{P}^{1-\sigma}}, \\ &= \frac{(z_{\omega})^{\delta} (p_{\omega})^{1-\sigma}}{\sum_{\omega \in \Omega} (z_{\omega})^{\delta} (p_{\omega})^{1-\sigma}}. \end{aligned} \tag{5}$$

Market shares play a crucial role for models based on the CES demand, as several key expressions can be rewritten in terms of them. Indeed, market shares can create a direct link between the model and the data.

3 CES as Representation of Differentiated Goods

The version of CES demand we presented allows for goods to exhibit both horizontal and vertical differentiation. Next, we analyze each form of differentiation separately.

3.1 Horizontal Differentiation

To simplify the explanation of horizontal differentiation, let's assume all varieties have equal appeal. In particular, suppose $z_{\omega} = 1$ for each $\omega \in \Omega$. When a good is horizontally differentiated, **each variety is perceived as unique** by the consumer. This means that no variety is considered superior or inferior to another—varieties are simply different from each other.

To characterize the agent's attitude towards these unique varieties, it is crucial that CES preferences exhibit strict convexity. When the number of varieties is endogenous, such property is referred to as **love for variety**. In simple terms, this property ensures consumer prefer diversifying consumption across varieties, rather than concentrating only on a strict subset of them. As a a corollary, any new variety introduced will be consumed.

This behavior is consistent with various real-world scenarios, such as those of individuals that vary their meals daily or prefer diversifying their clothing. This is in contrast to individuals that have strong preferences for a few specific meals or clothes, who may be reluctant to try new variants.

To show that the CES represents strictly convex preferences, we need to show that indifference curves are strictly convex.
Indifference curves are the combinations of goods that provide the same utility. To derive them, we use that

$$\mathrm{d}U = \sum_{\omega \in \Omega} \frac{\partial U}{\partial q_\omega} \mathrm{d}q_\omega$$

where $\frac{\partial U}{\partial q_\omega} = \left(\sum_{\omega \in \Omega} q_\omega^{\frac{\sigma - 1}{\sigma}} \right)^{\frac{\sigma}{\sigma - 1} - 1} q_\omega^{\frac{\sigma - 1}{\sigma} - 1}$.
Fix the utility, so that $\mathrm{d}U = 0$. Moreover, suppose that $\mathrm{d}q_{\omega'} \neq 0$ and $\mathrm{d}q_{\omega''} \neq 0$, with $\mathrm{d}q_\omega = 0$ for any $\omega \in \Omega \backslash \{\omega', \omega''\}$.
Then,

$$0 = \left[\left(\sum_{\omega \in \Omega} q_\omega^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1} - 1} q_{\omega'} \right] \mathrm{d}q_{\omega'} + \left[\left(\sum_{\omega \in \Omega} q_\omega^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1} - 1} q_{\omega''} \right] \mathrm{d}q_{\omega''}.$$

From this we get,

$$\frac{\mathrm{d}q_{\omega'}}{\mathrm{d}q_{\omega''}} = - \left(\frac{q_{\omega''}}{q_{\omega'}} \right)^{-\frac{1}{\sigma}},$$

and therefore $\frac{\mathrm{d}q_{\omega'}}{\mathrm{d}q_{\omega''}} < 0$.
Finally, indifference curves are convex since

$$\frac{\mathrm{d}^2 q_{\omega'}}{(\mathrm{d}q_{\omega''})^2} = \frac{1}{\sigma} \frac{1}{q_{\omega''}} \left(\frac{q_{\omega''}}{q_{\omega'}} \right)^{-\frac{1}{\sigma}} > 0.$$

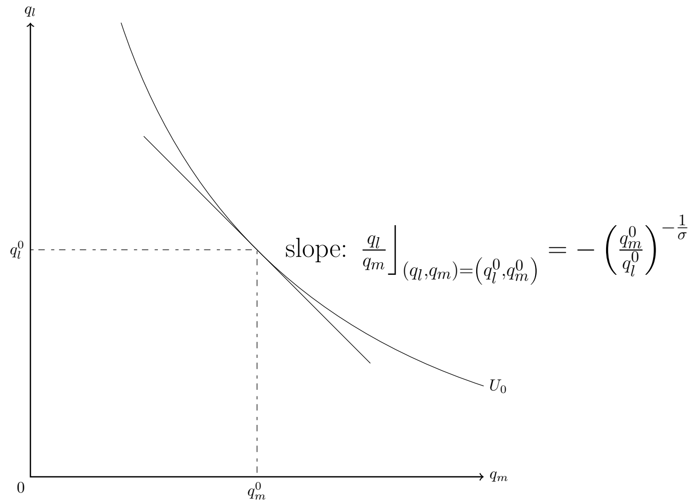
The intensity in which a consumer perceives varieties as different is governed by the elasticity of substitution, σ. This parameter measures the degree of substitution of a variety relative to the rest of varieties.

As the parameter σ is constant for all varieties, the degree of substitution remains the same, regardless of the number of available variety. This implies that we rule out crowding effects: when a firm enters an industry and introduces a new variety, the degree of substitution among the varieties is unchanged.

To formally show the role of σ, consider two varieties l and m. Indifference curves are then given by

$$\frac{\mathrm{d}q_l}{\mathrm{d}q_m} = -\left(\frac{q_m}{q_l}\right)^{-\frac{1}{\sigma}}.$$

Graphically:



In particular, the indifference curve when an agent does not consume the variety m (i.e., $q_m \to 0$) is

$$\lim_{q_m \to 0} \frac{\partial q_l}{\partial q_m} = -\infty.$$

The equation indicates that the consumer is willing to give up an infinitely large amount of variety l to consume a positive quantities of m. This implies that the agent finds any variety valuable, thus reflecting a preference for diversification. In particular, it determines that the consumer will always consume a new variety, regardless of the price charged.

The fact that the introduction of new varieties is valuable can be shown in an alternative way. Suppose that the agent is consuming an identical quantity $\widetilde{q}$ of each variety:

$$\begin{aligned} U_1 &= \left(\sum_{\omega\in\Omega} \widetilde{q}^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}, \\ &= M^{\frac{\sigma}{\sigma-1}}\widetilde{q}. \end{aligned}$$

Let's compare this to a scenario where the consumption per variety is halved, but the number of varieties is doubled. Formally, the number of varieties becomes $2M$ and the quantities per variety $\frac{\widetilde{q}}{2}$, so that

$$U_2 = (2M)^{\frac{\sigma}{\sigma-1}} \frac{\widetilde{q}}{2}.$$

Using that $\sigma > 1$, it is easy to see that

$$(2M)^{\frac{\sigma}{\sigma-1}} \frac{\widetilde{q}}{2} > M^{\frac{\sigma}{\sigma-1}} \widetilde{q},$$

and therefore $U_2 > U_1$. The result clearly illustrates the principle behind *love of variety*: the agent prefers a basket with a wider range of varieties, over one containing more of the original varieties.

3.2 Vertical Differentiation

When a good is vertically differentiated, the agent perceives some varieties as superior to others. The phenomenon typically arises when there are clear differences in quality or performance among varieties. For instance, this is the case with computers, where consumers generally prefer higher processing speeds, all else being equal.

The intensity in which a variety is preferred relative to the rest is captured by the parameter z_ω. Formally, (1) establishes that a higher z_ω yields more utility from consuming variety ω.

The fact z_ω enters directly into the utility function also explains why we referred to z_ω as appeal rather than simply quality: it encompasses all non-price aspects that impact a consumer's decision. As such, appeal represents not only objective features that make a variety superior, but also subjective psychological factors. For example, a consumer may prefer a particular brand due to its reputation, design, or perceived prestige, even if the objective quality is similar to other brands.

The incorporation of appeal allows the model to capture a wide range of consumer preferences and behaviors than if we only restrict to objective quality. Basically, we only suppose that consumer derives more utility from ω when z_ω is higher, remaining agnostic about the underlying reasons for this.

The terms $(z_\omega)_{\omega\in\Omega}$ capture vertical differentiation in relative terms, rather than absolute terms. This means that what matters for variety ω is z_ω in comparison to $(z_\omega)_{\Omega\setminus\omega}$. This can be easily observed by noting that any monotone transformation of the utility function still represents the same preferences. Hence, we could work instead with the following utility function

$$U\left[(q_\omega)_{\omega\in\Omega}\right] = \frac{\left[\sum_{\omega\in\Omega}(z_\omega)^{\frac{\delta}{\sigma}}(q_\omega)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}}{\left(\sum_{\omega'\in\Omega} z_{\omega'}\right)^{\frac{\delta}{\sigma}}}.$$

This means we can express the utility function through a normalized parameter $\widetilde{z}_\omega := \frac{z_\omega}{\sum_{\omega'\in\Omega} z_{\omega'}} \in (0,1)$, such that

$$U\left[(q_\omega)_{\omega\in\Omega}\right] := \left[\sum_{\omega\in\Omega}(\widetilde{z}_\omega)^{\frac{\delta}{\sigma}}(q_\omega)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}.$$

The degree to which vertical aspects influence consumption is governed by the parameter δ. This parameter directly influences the appeal elasticity of demand, which is given by

$$\frac{\mathrm{d}\ln q_\omega}{\mathrm{d}\ln z_\omega} = \delta\left(1-s_\omega\right).$$

Note that $\frac{\mathrm{d}\ln q_\omega}{\mathrm{d}\ln z_\omega} = \delta$ when firms have a negligible market share, as occurs in models of monopolistic competition.

We express q_ω as a function $q_\omega\left[p_\omega, z_\omega, \mathbb{P}\left((p_{\omega'}, z_{\omega'})_{\omega'\in\Omega}\right)\right]$. Taking logs of (3),

$$\ln q_\omega = \ln y + \delta\ln z_\omega - \sigma\ln p_\omega - (1-\sigma)\ln\mathbb{P},$$

and so

$$\frac{\mathrm{d}\ln q_\omega}{\mathrm{d}\ln z_\omega} = \delta - (1-\sigma)\frac{\partial\ln\mathbb{P}}{\partial\ln z_\omega}.$$

Taking logs of (4),

$$\ln\mathbb{P} = \frac{1}{1-\sigma}\ln\left[\sum_{\omega\in\Omega}(z_\omega)^{\delta}(p_\omega)^{1-\sigma}\right],$$

and taking the derivative with respect to z_ω,

$$\frac{\partial\ln\mathbb{P}}{\partial\ln z_\omega} = \frac{\delta}{1-\sigma}\frac{(z_\omega)^{\delta}(p_\omega)^{1-\sigma}}{\sum_{\omega\in\Omega}(z_\omega)^{\delta}(p_\omega)^{1-\sigma}}.$$

Finally, using (5), we determine that

$$\frac{\partial\ln\mathbb{P}}{\partial\ln z_\omega} = \frac{\delta}{1-\sigma}s_\omega,$$

and so $\frac{\mathrm{d}\ln q_\omega}{\mathrm{d}\ln z_\omega} = \delta\left(1-s_\omega\right)$.

In the literature, there are more specific versions of the CES that account for vertical differentiation. They can be understood as special cases by defining δ accordingly, implicitly assuming a given importance of appeal aspects for consumers.

One of these ways is by defining (1) with $\delta = \sigma - 1$, so that utility becomes

$$U\left[(q_\omega)_{\omega\in\Omega}\right] := \left[\sum_{\omega\in\Omega}(z_\omega q_\omega)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}.$$

Variety ω's optimal demand implies an expenditure on ω given by

$$r_\omega = y\frac{(p_\omega/z_\omega)^{1-\sigma}}{\mathbb{P}^{1-\sigma}},$$

where $\mathbb{P} := \left[\sum_{\omega\in\Omega}(p_\omega/z_\omega)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$.

This variant of the CES utility captures scenarios where consumers decide according to **the price per unit of quality provided by the variety**.

4 Price Elasticity of Demand

The price elasticity of variety ω is defined by

$$\varepsilon_\omega := -\frac{\mathrm{d}q_\omega/q_\omega}{\mathrm{d}p_\omega/p_\omega} := -\frac{\mathrm{d}\ln q_\omega}{\mathrm{d}\ln p_\omega},$$

where $q_\omega(p_\omega,\mathbb{P})$ is given by (3). Since $\mathbb{P}$ depends on p_ω due to (4), we get

$$\varepsilon_\omega := -\left[\frac{\partial\ln q_\omega}{\partial\ln p_\omega} + \frac{\partial\ln q_\omega}{\partial\ln\mathbb{P}}\frac{\partial\ln\mathbb{P}}{\partial\ln p_\omega}\right],$$

which equals

$$\varepsilon_\omega = \sigma - (\sigma-1)\,s_\omega. \tag{6}$$

Taking logs of (3),

$$\ln q_\omega = \ln y + \delta\ln z_\omega - \sigma\ln p_\omega - (1-\sigma)\ln\mathbb{P},$$

and so

$$\frac{\mathrm{d}\ln q_\omega}{\mathrm{d}\ln p_\omega} = -\sigma - (1-\sigma)\frac{\partial\ln\mathbb{P}}{\partial\ln p_\omega}.$$

Taking logs of (4),

$$\ln\mathbb{P} = \frac{1}{1-\sigma}\ln\left[\sum_{\omega\in\Omega}(z_\omega)^\delta(p_\omega)^{1-\sigma}\right],$$

and derivating with respect to p_ω,

$$\frac{\partial\ln\mathbb{P}}{\partial p_\omega} = \frac{1}{1-\sigma}\frac{(1-\sigma)(z_\omega)^\delta(p_\omega)^{-\sigma}}{\sum_{\omega\in\Omega}(z_\omega)^\delta(p_\omega)^{1-\sigma}},$$

$$\Rightarrow\frac{\partial\ln\mathbb{P}}{\partial\ln p_\omega} = \frac{(z_\omega)^\delta(p_\omega)^{1-\sigma}}{\sum_{\omega\in\Omega}(z_\omega)^\delta(p_\omega)^{1-\sigma}}.$$

Finally, using (5), we determine that

$$\frac{\partial\ln\mathbb{P}}{\partial\ln p_\omega} = s_\omega,$$

and the result follows.

Equation (6) implies that a firm with higher market share faces a less elastic demand. Thus, the CES parsimoniously captures market power through a firm's market share.

Likewise, (5) reveals that a lower price or a higher appeal commands a higher market share.

Note that s_ω affects the price elasticity through the term $\frac{\partial \ln \mathbb{P}}{\partial \ln p_\omega} = s_\omega$. Moreover, we know that the price index represents the aggregate conditions of the market. Both aspects entail that the strength of firm ω's impact on the industry conditions depends on its market share.

This fact has important implications for models with a continuum of varieties. In that case, the effect of ω's price on the price index is negligible, implying that

$$\varepsilon_\omega = \sigma$$

Consequently, *when the number of varieties is infinite, the price elasticity of demand equals the elasticity of substitution.*

5 Welfare

It can be shown that the CES utility function represents homothetic preferences, thereby exhibiting various additional properties. One of them is that the indirect utility function can be expressed as real income, $\frac{y}{\mathbb{P}}$.

Formally, this means that we can always define real numbers $\mathbb{Q}$ and $\mathbb{P}$, such that $\mathbb{Q}\mathbb{P} = y$ and where $\mathbb{Q}$ is the indirect utility function V. In the particular case of the CES, $\mathbb{Q}$ and $\mathbb{P}$ are given by

$$\mathbb{Q} := \left[\sum_{\omega \in \Omega} (z_\omega)^{\frac{\delta}{\sigma}} \, q_\omega^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}},$$

$$\mathbb{P} := \left[\sum_{\omega \in \Omega} (z_\omega)^{\delta} \, (p_\omega)^{1-\sigma} \right]^{\frac{1}{1-\sigma}},$$

where $\mathbb{Q}$ uses the optimal demands given by (3).

The fact that $\mathbb{Q}\mathbb{P} = y$ justifies why $\mathbb{Q}$ is referred to as a **quantity index** and $\mathbb{P}$ as a **price index**: $\mathbb{Q}$ can be interpreted as a representative basket of varieties, whereas $\mathbb{P}$ would be the necessary income to buy one unit of this basket.

$\mathbb{QP} = y$ additionally implies that

$$\mathbb{Q} = \frac{y}{\mathbb{P}},$$

where $\mathbb{Q}$ equals the consumer's indirect utility function. This establishes that one unit of utility corresponds to one unit of $\mathbb{Q}$, so that $V = 1$ is equivalent to $\mathbb{Q} = 1$. Based on this result, $\mathbb{P}$ also reflects the consumer's valuation of the basket, **as it constitutes the minimum expenditure that is necessary to achieve one unit of utility**. In formal terms, $\mathbb{Q} = 1$ when income is given by $y = \mathbb{P}$, implying that $V = 1$.[2]

We can prove directly the indirect utility function equals real income. The indirect utility function V corresponds to the utility function evaluated at the optimal quantities:

$V = \left[\sum_{\omega\in\Omega} (z_\omega)^{\frac{\delta}{\sigma}} (q_\omega)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}$

$\Rightarrow V = \left[\sum_{\omega\in\Omega} (z_\omega)^{\frac{\delta}{\sigma}} \left(\frac{(z_\omega)^{\delta}(p_\omega)^{-\sigma}}{\mathbb{P}^{1-\sigma}} y\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}$

$\Rightarrow V = \left[\sum_{\omega\in\Omega} (z_\omega)^{\frac{\delta}{\sigma}} \left(\frac{(z_\omega)^{\delta}(p_\omega)^{-\sigma}}{\mathbb{P}^{1-\sigma}} y\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}$

$\Rightarrow V = \left[\left(\frac{y}{\mathbb{P}^{1-\sigma}}\right)^{\frac{\sigma-1}{\sigma}} \sum_{\omega\in\Omega} (z_\omega)^{\delta} (p_\omega)^{1-\sigma}\right]^{\frac{\sigma}{\sigma-1}}$

$\Rightarrow V = \left(\frac{y}{\mathbb{P}^{1-\sigma}}\right) \left[\sum_{\omega\in\Omega} (z_\omega)^{\delta} (p_\omega)^{1-\sigma}\right]^{\frac{\sigma}{\sigma-1}}$

Given that $\mathbb{P} := \left[\sum_{\omega\in\Omega} (z_\omega)^{\delta} (p_\omega)^{1-\sigma}\right]^{\frac{1}{\sigma-1}}$ and hence $\left[\sum_{\omega\in\Omega} (z_\omega)^{\delta} (p_\omega)^{1-\sigma}\right]^{\frac{\sigma}{\sigma-1}} = \mathbb{P}^{-\sigma}$, then $V = \left(\frac{y}{\mathbb{P}^{1-\sigma}}\right) \mathbb{P}^{-\sigma} = \frac{y}{\mathbb{P}}$ and the result follows.

5.1 Welfare Determinants

To keep matters simple, suppose that the prices and quality of each variety are the same. Formally, let $p_\omega = \overline{p}$ and $z_\omega = \overline{z}$ for each ω. Then, the price index given by (4) is

$$\begin{aligned}\mathbb{P} &= \left[\sum_{\omega\in\Omega} (z_\omega)^{\delta} (p_\omega)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}, \\ &= M^{\frac{1}{1-\sigma}} (\overline{z}_\omega)^{\frac{\delta}{1-\sigma}} \overline{p}_\omega.\end{aligned}$$

Therefore, welfare is

$$\frac{y}{\mathbb{P}} = \mathbb{Q} = \frac{y}{M^{\frac{1}{1-\sigma}} (\overline{z}_\omega)^{\frac{\delta}{1-\sigma}} \overline{p}_\omega}, \tag{7}$$

where the powers of M and $\overline{z}_\omega$ are negative since we suppose $\sigma > 1$.

[2] The price index also plays another role. To see this, denote the Lagrange multiplier of the optimization problem (2) by λ^*. Remember that $\lambda^* := \frac{\partial V}{\partial y}$ due to the Envelope Theorem, and so λ^* is the marginal utility of income: the impact on optimal utility of one additional unit of income. The relation between λ^* arises since $\lambda^* = \frac{1}{\mathbb{P}}$, so that $\frac{1}{\mathbb{P}}$ is the marginal utility of income.

One unit of the quantity index (i.e., one basket of goods) represents one unit of real income, and hence one unit of utility. Likewise, the price index represents the income required to obtain one unit of utility, thereby representing the consumer's valuation of one unit of the basket. An implication of this is that **the lower the price index, the lower the income you need to obtain one unit of utility**. Consequently, **a lower price index reflects a higher consumer's valuation of the basket.** More generally, this implies that the factors affecting price index completely determine the valuation of the basket. Specifically, applying logarithms to the definition of the price index,

$$\mathbb{P} = \frac{1}{1-\sigma}\ln M + \frac{\delta}{1-\sigma}\ln \overline{z}_{\omega} + \ln \overline{p}_{\omega},$$

so that

- $\frac{\partial \ln \mathbb{P}}{\partial \ln M} < 0$: a higher number of varieties decreases the price index, leading to welfare improvements. This reflects that a basket comprising more varieties is more valuable, as it represents a more diversified basket.
- $\frac{\partial \ln \mathbb{P}}{\partial \ln \overline{z}_{\omega}} < 0$: a higher appeal of varieties decreases the price index, which increases welfare.
- $\frac{\partial \ln \mathbb{P}}{\partial \ln \overline{p}_{\omega}} > 0$: a higher price of varieties increases the price index, which reduces welfare.

Note that σ regulates the impact of the number of varieties on welfare, as it captures the intensity in which the consumer loves variety. Specifically, applying logs to $\mathbb{Q}$ and taking the derivative,

$$\frac{\partial \ln \mathbb{Q}}{\partial \ln M} = \frac{1}{\sigma - 1}.$$

This establishes that the impact of M on $\mathbb{Q}$ is lower when σ is higher. The outcome reflects that consumer perceives varieties as less differentiated when σ is higher. In the limit, where $\sigma \to \infty$, the utility function becomes linear, representing a scenario where varieties are seen as perfect substitutes. This explains why $\left.\frac{\partial \ln \mathbb{Q}}{\partial \ln M}\right\rfloor_{\sigma\to\infty} = 0$, as consumption diversification has no value when varieties are perfect substitutes—the consumer's sole concern is total consumption, without caring about whether one or multiple varieties are consumed.

6 A Continuum of Goods

While we have assumed a discrete number M of varieties, it is standard to work with a continuum of varieties. This entails that the number of varieties is infinite, with every point in the interval $[0, M]$ representing a different variety.[3]

The assumption is in particularly adopted in models of monopolistic competition, implying that every variety is negligible for an industry's aggregate conditions. In terms of the CES, this is captured by saying that no firm in isolation is capable of affecting the price index.

Formally, the utility function is

$$U := \left[\int_0^M (z_\omega)^{\frac{\delta}{\sigma}} (q_\omega)^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}\omega\right]^{\frac{\sigma}{\sigma-1}}.$$

The optimization problem is now

$$\max_{(q_\omega)_{\omega\in[0,M]}} U := \left(\int_0^M (z_\omega)^{\frac{\delta}{\sigma}} (q_\omega)^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}\omega\right)^{\frac{\sigma}{\sigma-1}} \text{subject to } y = \int_0^M p_\omega q_\omega \, \mathrm{d}\omega,$$

with the same solution as before

$$q_\omega = y\frac{(z_\omega)^\delta (p_\omega)^{-\sigma}}{\mathbb{P}^{1-\sigma}},$$

but with the difference that the price index in the continuum version is

$$\mathbb{P} := \left[\int_0^M (z_\omega)^\delta (p_\omega)^{1-\sigma} \, \mathrm{d}\omega\right]^{\frac{1}{1-\sigma}}.$$

Likewise, the indirect utility function is the same as in the discrete case:

$$V(\mathbb{P}) = \frac{y}{\mathbb{P}}.$$

The optimization problem can be obtain through the solution for the discrete case, which we then extend to a continuum of goods. Alternatively, we can use either calculus of variation or optimal control.

Regarding calculus of variation, consider the solution q_ω and an additive perturbation δq_ω. This implies that the Lagrangean can be expressed in terms of $q_\omega + \delta q_\omega$ by,

$\mathscr{L} := \int_0^M (z_\omega)^{\frac{\delta}{\sigma}} [q_\omega + \delta q_\omega]^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}\omega + \lambda\left(Y - \int_0^M p_\omega [q_\omega + \delta q_\omega] \, \mathrm{d}\omega\right)$

$\frac{\partial \mathscr{L}}{\partial \delta} = \frac{\sigma-1}{\sigma} (z_\omega)^{\frac{\delta}{\sigma}} [q_\omega + \delta q_\omega]^{\frac{\sigma-1}{\sigma}-1} q_\omega - \lambda q_\omega p_\omega = 0$

[3] It is common to say that there is as a continuum of goods with measure M. This is analogous to say that in the interval $[0, 1]$ there are an infinite number of goods, but the length of the line is equal to 1 and so the measure of goods equals one.

Evaluating the solution at $\delta = 0$,
$\Rightarrow \frac{\partial \mathscr{L}}{\partial \delta}\Big|_{\delta=0} = \frac{\sigma-1}{\sigma}(z_\omega)^{\frac{\delta}{\sigma}}[q_\omega]^{\frac{\sigma-1}{\sigma}-1} q_\omega - \lambda q_\omega p_\omega = 0$
$\Rightarrow \frac{\partial \mathscr{L}}{\partial \delta}\Big|_{\delta=0} = \frac{\sigma-1}{\sigma}(z_\omega)^{\frac{\delta}{\sigma}}[q_\omega]^{\frac{\sigma-1}{\sigma}-1} - \lambda p_\omega = 0$ for all $\omega \in [0, M]$
which gives the same expression as the first-order condition we derived.

Consider now an optimal-control approach. Suppose we define an auxiliary variable $Y^R(\omega)$, which is the residual income after consuming good ω. The variable Y^R is constrained to $Y^R(0) = Y$ and $Y^R(M) = 0$. Hence, we can express the control variable as $\frac{\mathrm{d}Y^R}{\mathrm{d}\omega} = -p_\omega q_\omega$, since each good reduces the residual demand. The Hamiltonian is
$\mathscr{H} := (z_\omega)^{\frac{\delta}{\sigma}}(q_\omega)^{\frac{\sigma-1}{\sigma}} + \lambda(-p_\omega q_\omega)$
So taking the first order conditions
$\frac{\partial \mathscr{H}}{\partial q_\omega} = \frac{\sigma-1}{\sigma}(z_\omega)^{\frac{\delta}{\sigma}}(q_\omega)^{\frac{\sigma-1}{\sigma}-1} - \lambda p_\omega = 0$ for each $\omega \in [0, M]$,
which provides the same solution.

One convenient feature of the continuum case is that no firm in isolation can influence aggregate variables. For instance, the price elasticity of demand for ω is

$$\varepsilon_\omega := -\left[\frac{\partial \ln q_\omega}{\partial \ln p_\omega} + \frac{\partial \ln q_\omega}{\partial \ln \mathbb{P}}\frac{\partial \ln \mathbb{P}}{\partial \ln p_\omega}\right].$$

But, since each firm is negligible for the aggregate conditions, $\frac{\partial \ln \mathbb{P}}{\partial \ln p_\omega} = 0$ and therefore

$$\varepsilon_\omega = \sigma.$$

7 CES from a Random Utility Model (OPTIONAL)

Our previous derivation of the CES demand relied on the assumption of a representative consumer with love for variety. However, this is not the only approach to derive a CES demand. Moreover, each of these approaches gives rise to alternative interpretations of the demand parameters.

In particular, we show next that the CES demand can also emerge from a random utility model, where the desire for diversification stems from an *ideal-variety* interpretation. In this framework, consumers are heterogeneous in their preferences, which leads them to favor different varieties in equilibrium. As a result, aggregate demand reflects a diversified pattern of consumption.

A useful illustration is the market for football jerseys. Suppose we segment a country's population according to their preferred team (for example, Barcelona, Real Madrid, or Sevilla). Each individual demands the jersey of their chosen team, and when these demands are aggregated, all jerseys are consumed. The outcome is diversified aggregate demand, even though each consumer purchases only one variety.

To formalize this idea, consider a continuum of agents with mass L, and a set of varieties $\Omega := \{1, 2, ..., N\}$. Each consumer makes two choices: which variety to consume and the quantities of it. Unlike the representative-consumer approach, consumers are **constrained to buy only one variety**, although they can buy as much as they wish of the variety chosen. This feature explains why the particular random-utility model considered is known as involving discrete-continuous choices.

The population of consumers exhibits heterogeneous preferences for each variety. This is formalized by an indirect utility of variety ω given by

$$V(y, p_\omega) := \ln y - \ln p_\omega + \underline{\varepsilon}_\omega \text{ for each } \omega \in \Omega,$$

where $\underline{\varepsilon}_\omega$ is a random variable. We suppose that each $\underline{\varepsilon}_\omega$ follows an iid Gumbel distribution, with standard deviation $\mu\frac{\pi}{6}$ where $\mu > 0$.[4] Note that a higher value for μ increases the variance, and hence represents greater dispersion of tastes within the population. This is why μ reflects the degree of heterogeneity in preferences.

Conditional on choosing ω, the quantity consumed can be obtained by Roy's identity: $q_\omega = -\frac{\partial V(y,p_\omega)/\partial p_\omega}{\partial V(y,p_\omega)/y}$, implying that

$$q_\omega(p_\omega, y) = \frac{y}{p_\omega}.$$

Once we determine the quantity consumed for each ω, we need to determine which variety will be chosen. To do this, let $\alpha_\omega(\mathbf{p})$ denote the proportion of consumers selecting variety ω when prices are $\mathbf{p}$. This term also corresponds to the probability of ω yielding the highest indirect utility. Formally,

$$\alpha_\omega(\mathbf{p}) := \Pr\left[\left(\ln y - \ln p_\omega + \mu\underline{\varepsilon}_\omega \geq \ln y - \ln p_{\omega'} + \mu\underline{\varepsilon}_{\omega'}\right)(\forall\omega' \in \Omega)\right].$$

By properties of the Gumbel distribution, $\alpha_\omega(\mathbf{p})$ has the following closed-form solution:

$$\alpha_\omega(\mathbf{p}) = \frac{\exp\left(\frac{\ln p_\omega}{\mu}\right)}{\sum_{\omega'\in\Omega}\exp\left(\frac{\ln p_{\omega'}}{\mu}\right)} \Rightarrow \alpha_\omega(\mathbf{p}) = \frac{(p_\omega)^{-\frac{1}{\mu}}}{\sum_{\omega'\in\Omega}(p_{\omega'})^{-\frac{1}{\mu}}}.$$

[4] The results we show would have been identical if we had defined the indirect utility function by $V(y, p_\omega) := \ln y - \ln p_\omega + \mu\underline{\varepsilon}_\omega$ and assume that $\underline{\varepsilon}_\omega$ is iid and with a Gumbel distribution that has zero mean and unit variance.

Thus, the aggregate demand is defined by

$$Q_{\omega}\left(\mathbf{p},y\right):=L\alpha_{\omega}\left(\mathbf{p}\right)q_{\omega}\left(p_{\omega},y\right),$$

which equals

$$Q_{\omega}\left(\mathbf{p},y\right)=L\frac{\left(p_{\omega}\right)^{-\frac{1}{\mu}}}{\sum_{\omega'\in\Omega}\left(p_{\omega'}\right)^{-\frac{1}{\mu}}}\frac{y}{p_{\omega}}\Rightarrow Q_{\omega}\left(\mathbf{p},y\right)=Ly\frac{\left(p_{\omega}\right)^{-\frac{1}{\mu}-1}}{\sum_{\omega'\in\Omega}\left(p_{\omega'}\right)^{-\frac{1}{\mu}}}.$$

A direct link to the CES demand can be established by defining $\sigma:=\frac{1+\mu}{\mu}$, which determines $Q_{\omega}\left(\mathbf{p},y\right)=Ly\frac{\left(p_{\omega}\right)^{-\sigma}}{\sum_{\omega'\in\Omega}\left(p_{\omega'}\right)^{1-\sigma}}$. In this model, σ does not correspond to the elasticity of substitution. Instead, μ measures the degree of taste heterogeneity within the population. In particular, $\frac{\partial\sigma}{\partial\mu}>0$, so that a higher σ reflects less heterogeneity.